

Bariscan Kurtkaya

RESEARCH INTEREST	I investigate how neural networks leverage low-dimensional manifold geometry in latent space to implement core cognitive capabilities—such as memorization, generalization, and comparison—and how this geometric structure helps us interpret what a model " <u>thinks</u> " while performing a given " <u>behaviour</u> ".		
EDUCATION	University of California, Santa Barbara , PhD in Electrical and Computer Eng., California, USA 2025 - Present Awarded PhD Fellowship from UCSB ECE Department . <i>Principal Investigator:</i> Prof. Nina Miolane Koc University , MSc in Computer Science and Engineering, Istanbul, Turkey 2025 Awarded KUIS AI Fellowship . <i>Principal Investigator:</i> Prof. Yucel Yemez <i>Thesis:</i> Understanding dynamical phases of short-term mechanisms Yildiz Technical University , BSc in Electronics and Communication Engineering, Istanbul, Turkey 2023 Graduated with High Honors .		
SELECTED PUBLICATIONS * FOR EQUAL CONTRIBUTION † FOR EQUAL SUPERVISION	[1] <i>A ghost mechanism: An analytical model of abrupt learning</i>, Physical Review X, 2026, [Paper] Bariscan Kurtkaya*, Fatih Dinc*, Ege Cirakman*, Mert Yuksekgonul, Yiqi Jiang, Mark J. Schnitzer†, Hidenori Tanaka† [2] <i>Dynamical phases of short-term memory mechanisms in RNNs</i>, ICML'25, 2025, [Paper] Bariscan Kurtkaya*, Fatih Dinc*, Mert Yuksekgonul, Marta Blanco-Pozo, Ege Cirakman, Mark J. Schnitzer, Yucel Yemez, Hidenori Tanaka†, Peng Yuan†, Nina Miolane† [3] <i>RAVE: Randomized Noise Shuffling for Fast and Consistent Video Editing with Diffusion Models</i>, CVPR'24 Highlight, 2024 [Paper] [Code] [Website] Bariscan Kurtkaya*, Ozgur Kara*, Hidir Yesiltepe, James M. Rehg, Pinar Yanardag†		
RECENT PREPRINTS	[1] <i>RNNs with equivalent attractor landscapes can implement distinct neural algorithms</i>, Under review in top ML Conference Bariscan Kurtkaya*, Mehmet Harmanli*, Peng Yuan, Yucel Yemez, Fatih Dinc†, Nina Miolane† [2] <i>Multimodal alignments of in vivo imaging and spatial biology datasets at cellular resolution</i>, Under review in top Neuroscience Journal [Paper] Lun Wang*, Xiqian Jiang*, Xiaochen Sun*, Gaurav Chattree, Ali Cetin, Xiaochun Cai, Elijah Paul, Radoslaw Chrapkiewicz, Oscar Hernandez, Yuxi Ke, Tomohisa Yoda, Fatih Dinc, Bariscan Kurtkaya, Yanping Zhang, Zhengji Zhang, Mark J. Schnitzer [3] <i>A theory of parameter identifiability in data-constrained recurrent neural networks</i>, Under review in top ML Conference Fatih Dinc, Abby Bertics, Bariscan Kurtkaya, Nina Miolane		
ACADEMIC HONORS & AWARDS	• PhD Fellowship, UC Santa Barbara ECE department 2025-2026 • Funded by the Feldman-McClelland Open-a-Door fund. 2024 • Impact Scholar, awarded a fellowship from the Bridge to Turkiye Fund. 2024 • KUIS AI Fellowship, Koc University AI Center 2023-2025 • 2209 TUBITAK Research Project Grant Holder, awarded from TUBITAK. 2022 • High Honors, Yildiz Technical University 2023		
RESEARCH EXPERIENCE	University of California, Santa Barbara , Geometric Intelligence Lab , Graduate Student Researcher 2025 - Present Supervised by Prof. Nina Miolane • Exploring how memory and intelligence emerge in neural networks by developing mechanistic interpretability methods grounded in neuroscience principles and dynamical systems theory, advancing the foundation for safer and more transparent AI systems. • Analyzing memory representations and dynamics across diverse architectures (Transformers, LSTMs, GRUs) using mechanistic interpretability tools, and collaborating with experimental neuroscience labs to validate computational findings in biological neural recordings. Stanford University , Schnitzer Group , Visiting Graduate Student Researcher 2024 - 2025 Supervised by Prof. Mark J. Schnitzer & Dr. Fatih Dinc • Investigated how short-term memory mechanisms emerge in recurrent neural networks by training biologically-plausible leaky firing-rate RNNs on memory tasks. Discovered evidence for sequential activity via limit cycles that are ex-		

perimentally indistinguishable from slow-point manifolds during trial periods. Developed analytical toy models and low-rank RNNs to characterize when and why different memory mechanisms emerge. **Published in ICML'25.**

- Gained hands-on neuroscience experience by participating in mice surgeries, optogenetics experiments, and two-photon Ca^{2+} imaging sessions. Contributed to neuron segmentation work using deep learning (Cellpose), resulting in co-authorship on a manuscript.

Koc University , [Koc University AI Center](#), Graduate Student Researcher

2023 - 2025

Supervised by [Prof. Yucel Yemez](#)

- Investigated whether RNNs employ short-term versus working-memory computations in delayed discrimination tasks by developing mechanistic interpretability tools to quantify information preservation. Trained inverse layers to reconstruct input signals from hidden states across diverse training conditions, revealing that RNNs consistently demonstrate short-term memory-like computation by preserving complete input information through delay periods. **Published in NeurIPS'25 NeurReps Workshop.**
- Identified learning-rate collapse as a fundamental barrier to training large-scale RNNs for computational neuroscience, showing that neuron count increases require proportional learning rate decreases (N^{-1}). Proposed utilizing low-rank RNN architectures to enable efficient training of networks with 100k+ neurons. **Published in NeurIPS'25 NeurReps Workshop.**

Virginia Tech, [GemLab VT](#), Graduate Researcher Intern

2023 - 2024

Supervised by [Prof. Pinar Yanardag](#)

- Developed a novel, training-free, zero-shot video editing framework, leveraging pre-trained text-to-image diffusion models to enable efficient video editing without additional training requirements. **Published in CVPR'24 Highlight.**
- Achieved state-of-the-art performance in both qualitative and quantitative evaluations, demonstrating superior editing capabilities and addressing the shape morphing problem, a limitation of previous methods.

WashU St. Louis, [McDonnell Center](#), Undergraduate Researcher

2022 - 2023

Supervised by [Prof. Tansu Daylan](#)

- Conducted research to enhance exoplanet discovery through machine learning on observations from the James Webb Space Telescope.
- Developed a physics-informed, GPU-accelerated data augmentation technique, overcoming data scarcity by expanding the dataset from 1000 to 10^6 in 10 minutes thereby enhancing model training capacity.

TEACHING EXPERIENCE

ECE3 - Python Programming for Science and Eng - UC Santa Barbara - Graduate Teaching Assistant

2025 Fall

- Delivered lab sessions on Python programming and Introduction to Data Science. Provided support during office hours, addressing student questions, and assessed student performance through grading course homeworks.

COMP 410/510 - Computer Graphics - Koc University - Graduate Teaching Assistant

2024,2025 Spring

- Delivered a lecture on 3D reconstruction using Neural Radiance Fields, introducing key concepts and applications to the class. Provided support during office hours, addressing student questions, and assessed student performance through grading course homeworks.

COMP 106 - Discrete Mathematics for CSE - Koc University - Graduate Teaching Assistant

2023 Fall

- Led problem solving sessions, presenting example problems to the class on the RSA public-key cryptosystem to enhance understanding. Provided support during office hours, addressing student questions, and assessed student performance through grading midterm and final exams.

TALKS

Voxel51 CVPR2024 Meetup - Speaker

2024 Summer

- Presented our CVPR work [RAVE: Randomized Noise Shuffling for Fast and Consistent Video Editing with Diffusion Models](#) at Voxel51's CVPR meetup.

Akbank MultiGroup - Women in AI - Speaker

2023 Spring

- Delivered an advanced-level seminar on deep learning-based object detection models to underrepresented groups at the Akbank MultiGroup MoreThan101 series. ([Presentation Link](#))

University of Milan - Visiting Research Intern

2022 Summer

- Presented a seminar on Capsule Networks to faculty and graduate students in the Computer Science Department at the University of Milan. ([Presentation Link](#))

EXTRACURRICULAR

Awarded 1st and 3rd place as in the US-based [MateRov](#) Semi-autonomous Underwater Systems competition organized under the sponsorship of [NSF](#), in consecutive years.

- Awarded 4th place in the Undersecretariat of Defense Industry award in the Underwater Systems category.
- Awarded 3rd place in the Teknofest Turkiye award in the Underwater Systems category.
- Hugging Face and Microsoft Ambassador in Turkey
- Nike and Adidas Pacer Athlete in Turkey
- Pro volleyball player in Turkey Volleyball League
- Volleyball player in Yildiz Technical University Team
- Amateur performer (covered by top platforms such as [Tepkikolik Channel](#) - 2.54 Million Subscribers)